

DAM501 – DATA MINING

Final Project Description

In this assignment, you will explore the application of data mining methods to solve a real-world business problem. Your task is to select a relevant dataset, apply data mining pipeline on this dataset, and highlighting the importance of summarizing the key findings and analyzing the overall project process. This will help you understand the strengths and weaknesses of different data mining techniques in a business context, and how to choose the right method for a particular task.

Instructions:

1. **Problem Definition (15%):**

- **Business Context:** Provide background on the business or industry that your dataset is related to, and describe the specific business challenge or opportunity you aim to address using data mining.
 - ✓ **Example:** In the retail industry, predicting customer churn can help businesses develop retention strategies and reduce customer turnover.
- **Objective of the ML Model:** Clearly define the goal of your model. Specify whether it is a **regression**, **classification**, or **clustering** problem, and justify why this type of ML approach is suitable.
 - ✓ **Example:** "The goal is to predict which customers are most likely to churn within the next three months based on their past behavior, demographics, and interaction data."
- **Business Impact:** Explain how solving this problem with machine learning could benefit the business. Mention potential gains, such as **increased revenue**, **cost savings**, or **operational efficiency**.
- **Assumptions and Limitations:** List any assumptions you are making about the data or the business problem, and mention any limitations.
- **Key Metrics for Success:** Define the key performance indicators (KPIs) or metrics that will determine if your model successfully solves the problem.

2. **Dataset Selection (5%):**

- Choose a real-world dataset relevant to your problem. The dataset should include input features (independent variables) and a target variable (dependent variable).
- Use public datasets from sources such as Kaggle, UCI Machine Learning Repository, or government databases.

3. **Data pre-processing (10%):** Show steps/techniques (e.g. handle missing values, outliers, etc.) suitable to pre-process the data, improve the data quality to make it ready for further processing.

4. **Exploratory Data Analysis (10%):**

- Conduct an exploratory analysis to understand the data and visualize important trends or patterns.
- Identify which features are likely to be most important for prediction.
- Summarize key insights from the data that may affect the choice of machine learning models.

5. **Model Selection (30%):**

- Choose at least **two** different machine learning algorithms such as:
 - **Linear models** (e.g., Linear Regression, Logistic Regression)
 - **Decision trees and ensemble models** (e.g., Decision Trees, Random Forest, Gradient Boosting, XGBoost)
 - **Support vector machines** (SVM)
 - **Neural networks** (e.g., Feedforward Neural Networks)
 - **Clustering algorithms** (if your problem involves unsupervised learning, e.g., K-Means)
 - **Apriori for analyzing frequent patterns**
 - **etc.**
- Implement each model using a machine learning framework like **Scikit-learn**, **TensorFlow**, or **PyTorch**.
- **For each model**, explain:
 - Why you chose that particular model for the problem.
 - How the model works conceptually (the key intuition behind the algorithm).
 - How you tuned hyperparameters (e.g., cross-validation, grid search, or random search).

6. **Model Evaluation and Comparison (20%):**

- Evaluate the performance of each model using appropriate metric.
- Discuss the strengths and weaknesses of each model based on:
 - Accuracy/performance.
 - Computation time/efficiency.
 - Interpretability, etc.

7. **Business Impact and Insights (10%):**

- Based on the results of your model comparison, recommend which machine learning algorithm would be best suited to solve the business problem and why.

- Explain how the chosen model can be deployed in a real-world business environment (e.g., customer retention system, sales forecasting, etc.).
- Provide insights from the model that could drive actionable business decisions (e.g., which customer segments are most likely to churn, or what factors influence sales the most).

8. Bonus: Model Optimization (Optional, +10%):

- Use techniques such as **feature engineering**, **dimensionality reduction** (e.g., PCA), or **ensemble learning** (e.g., stacking, boosting) to improve the performance of your best model.
- Compare the optimized model's performance with the original models.

Deliverables:

1. Project Report (PDF or Word Document):

- Clearly structured sections:
 1. Introduction and Dataset Description
 2. Problem Definition
 3. Data Pre-processing
 4. Exploratory Data Analysis
 5. Model Implementation and Results
 6. Model Comparison
 7. Business Impact and Recommendations
- Include visualizations (graphs, tables, plots) where necessary.

2. Code Submission (Jupyter Notebook or Python Script):

- Clean and well-commented code implementing the models.
- Ensure that your code is reproducible (include environment details if necessary, e.g., versions of libraries used).

3. Presentation:

- Prepare a short presentation summarizing your findings, model comparison, and business insights.